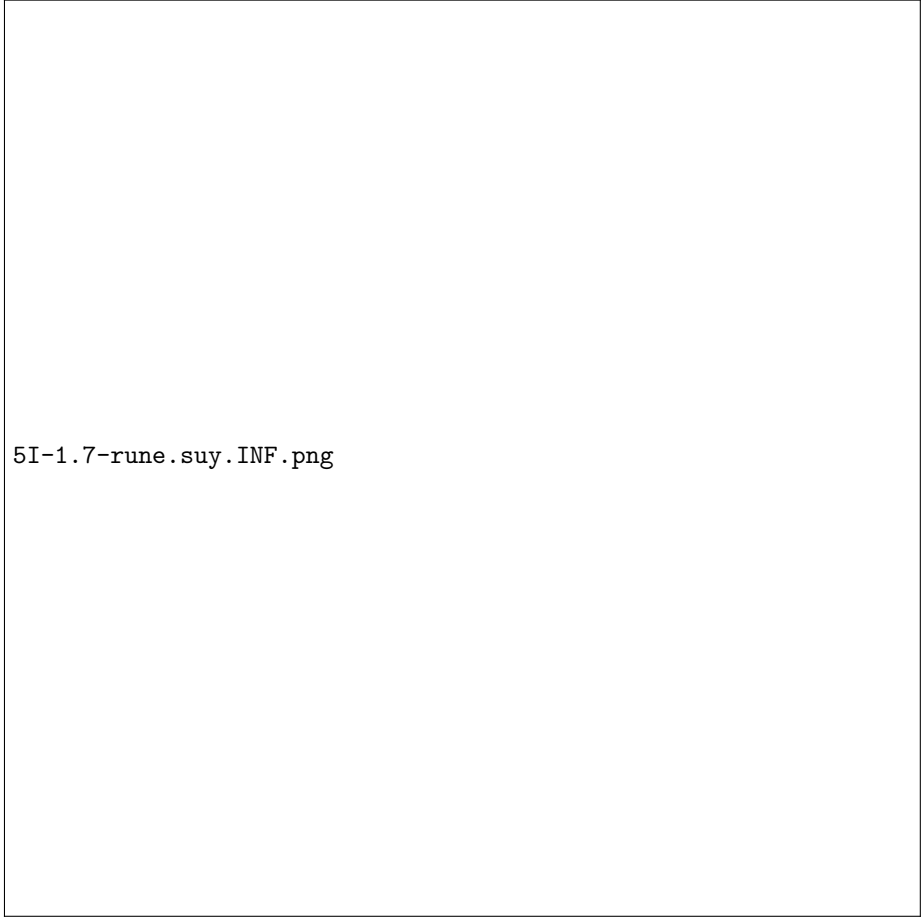


Auteur: Rune Suy
Studierichting: Informatica
Oefeningenreeks 5I nr. 1.7

$$\begin{aligned}f(x) &= x^3 + \frac{1}{12x} = \frac{12x^4 + 1}{12x} \text{ over } [1, 3] \\f'(x) &= \frac{12x \cdot 48x^3 - (12x^4 + 1) \cdot 12}{144x^2} \\&= \frac{36x^4 - 1}{12x^2} \\f'(x)^2 &= \frac{1296x^8 - 72x^4 + 1}{144x^2} \\1 + f'(x)^2 &= \frac{1296x^8 + 72x^4 + 1}{144x^2} = \left(\frac{36x^4 + 1}{12x^2} \right)^2 \\&= \int_1^3 \sqrt{\left(\frac{36x^4 + 1}{12x^2} \right)^2} dx = \int_1^3 \left(\frac{36x^4 + 1}{12x^2} \right) dx \\&= 3 \left[\frac{x^3}{3} \right]_1^3 + \frac{1}{12} \left[\frac{-1}{x} \right]_1^3 \\&= \frac{469}{18}\end{aligned}$$



5I-1.7-rune.suy.INF.png

References